StimmFolgen

Technical report - Hack Apertus Online, Track 2B - 15 October 2026

1. A private vote tracker with a reviewed historical case

StimmFolgen helps a voter examine what a referendum could change, who might be affected and what its source material does not establish. It gives no voting recommendation. The implemented prototype covers the 13th AHV old-age pension initiative of 3 March 2024 using the original German Federal Chancellery booklet.

The responsive desktop/mobile journey has three simple entry points: My votes, Upcoming votes and Search. A reader tracks a proposal without implying a Yes/No choice, then opens Explore my vote for its meaning, dated sources, official updates, related proposals and Me/Others/Switzerland perspectives. Detailed evidence is progressively disclosed. The catalogue contains five selected federal proposals; only AHV has reviewed Apertus analysis.

The October 2026 official AHV implementation update and November financing relationship were manually verified against official publisher pages. They are separate from the original pre-vote analysis, and not new model outputs. Source publication/check dates and explicit no-recorded-update states prevent historical forecasts being mistaken for observed outcomes. These web excerpts carry text hashes, not original HTML-byte provenance.

The overview is a **reviewed recorded case**, with wording accepted after real local Apertus inference and independent checks. The public Ask Apertus form now calls live Public AI Apertus v1.5 8B through Cloudflare. Fresh results are explicitly AI-generated and not human reviewed; source IDs resolve from authenticated evidence, and a separate same-model audit can withhold an answer. The five-year cost statement is labelled reviewed source-reference context; it was not extracted in the final happy-path model trial.

Bounded Gate 0 was cleared before frontend files were created. This is a narrow, tested vertical slice, not a production civic-information service or a general policy accuracy claim.

2. Evidence pipeline and publication boundary

Original PDF bytes are authenticated, extracted with PyMuPDF 1.28.2 and indexed as 91 immutable source spans. Apertus selects span IDs and typed fields; quotations resolve from the ledger rather than being generated. Every selected ID must resolve to the source edition and correct PDF page. A separate, evaluation-only bounded semantic reference checks predicate, values, actor, epistemic type and missing facts. It is coding-agent-authored, not an independent academic benchmark. The user reviewed the five published statement meanings and wording, not every raw control trace.

Actor attribution uses two real completions: neutral source evidence/actor extraction and independent parsing of the requested actor. Application logic compares these actors symmetrically and retains extracted IDs unchanged. Neither model prompt nor comparator receives the expected actor, gold span or evaluation reference. The resulting conflict/refusal is application-composed, not a single LLM-generated answer.

The source hash is 53c7cc72c28f80969ed56d7813ad8c05e505b1817923418d0dfa45368a6bd07e (759,783 bytes, 32 pages). Its printed content cutoff is 29 November 2023. An exact-byte archived official copy establishes availability by 11 January 2024, 12:11:17 UTC. This is an archive-observation upper bound, not the first publication instant or a signed timestamp. The public case frame is 2 March 2024, 23:59:59 CET. These dates remain separate; older traces/span identities are not rewritten.

At backend startup, the release guard authenticates the original packet and release manifest, re-extracts the catalogue, checks pinned recorded model identity, verifies temporal admission, requires five bound ACCEPT decisions and independently replays the controls. Missing approval, changed source/model or failed control blocks startup. New inference candidates cannot automatically replace reviewed published statements.

The smaller implemented runtime contract is documented in docs/APP_CONTRACT.md. The larger blueprint remains a design target. ADR 0006 documents the deliberately smaller implemented browser-vault contract and its departures from that blueprint.

3. Real local Apertus use

Inference ran on the user's Apple Silicon M3 Pro Mac with 18 GiB RAM, macOS 26.6.2 arm64. Existing assets were inspected before runtime selection. The successful weights are a community text-only MLX 4-bit conversion, tokimoa/apertus-v1.5-8b-mlx-4bit, revision f08ed532a657bef7d322bd0cbca43dac13a9d0a3. Declared upstream is swiss-ai/Apertus-v1.5-8B, revision a411d838600baf0e3635a3daf66fb7c55fc97bb6. Model-card metadata declares Apache-2.0. Independent equivalence to official BF16 weights is not tested; multimodal capability is not claimed.

The safetensors weights occupy 4,530,491,646 bytes; SHA-256 is d0478edeb06bf18e5f4f7b50db77d89b839730b2d16cb1020015e6099e42074b. Group size 64, affine quantisation, 4 bits. Runtime: MLX 0.32.0, mlx-lm 0.31.3, transformers 5.15.0. The loopback server advertises the verified local path and serves default_model. Hub network access and telemetry were disabled for these local runs; they used no inference API.

Actual requests use temperature 0, 500 completion tokens and thinking disabled. The fresh 14 October release suite made eight genuine model completions for seven trials: provision, introduction cost, financing refusal, fabricated quote, false actor attribution, unsupported salary-based CHF charge and later-knowledge leakage. All seven passed independent bounded controls. The three-response combined happy path passed but remains an aggregate, not another completion.

Observed trial latency was 6.90-31.76 seconds, with the two-call actor trial taking 19.39 seconds. These timings measure this local execution, not production throughput or a service guarantee. Raw requests, HTTP responses, candidate fields, hashes, model metadata, timings and reference snapshots are preserved under docs/evidence/gate0-release-2026-10-14. Only the absolute local model directory is redacted. All preceding failed traces remain intact.

4. Evaluation, limitations and privacy

The earlier full-booklet direct Apertus baseline ran genuinely and failed mechanical controls, including unsupported financing, required coverage and quote/role checks; latency was 54.95 seconds. The extractive baseline returns three real source passages with no model interpretation. Earlier generated-quote and span-selection protocols failed reproducibly. In particular, the source-actor discovery fix could select the government then relabel the answer as committee; immutable selection plus independent actor comparison removed that failure in the bounded case.

These are development diagnostics with different protocols, not a controlled held-out accuracy comparison. A quote check or schema pass cannot prove meaning. The current seven-trial result is pipeline performance on one AHV development case; retrieval, task types, null personal costs and refusal policy are application-supplied. The user approved P01-P05 source statements. No human approval of T01-T07 raw trials is claimed.

One hundred deterministic Python tests pass locally, covering source/span integrity, independent semantic rejection, actor comparison, public timing, pinned identity, review bindings, HTTP routing and privacy rejection. Separate Docker smoke checks load the real case, exercise personal-cost and unsupported-question abstention, reject private extra fields and serve authenticated source bytes. Browser checks cover desktop/mobile tracking, local search, source dates, keyboard tabs, Me/Others/Switzerland, honest unavailable analysis, encrypted backup download/recovery and reload locking. Native WebCrypto tests additionally reject wrong passwords/tampering and check fresh nonces, concurrent writes and cancelled lock races. No user comprehension study or comprehensive accessibility audit has been run.

Visit-only tracking is the default. Persistent choices and notes are opt-in: a passphrase derives a nonextractable AES-GCM-256 key with PBKDF2-SHA256 (600,000 iterations, random 16-byte salt); every write uses a fresh 12-byte nonce. Only authenticated ciphertext and cryptographic parameters persist. Keys stay in memory; reload, page hiding, idle timeout and cross-tab changes lock the private view. Backups are encrypted and restored locally. No passphrase recovery or cross-device sync exists.

Private choices, notes, local search and perspective selections are not attached to backend/model requests. A prior recorded-case journey captured 33 public GETs and zero POSTs; the live service now sends only the explicit public question in POST requests. That earlier count is not evidence of the live network flow. The production server does not log requests. Same-origin CSP, no-store responses and rejection of extra private API fields remain. External official-source visits can be observed by their publishers. Encryption does not protect an unlocked or compromised browser/device; this implementation has not received an independent security audit.

Unverified: second referendum, unseen-ballot generalisation, language parity, independently audited production security, current-law coverage and official-BF16 conversion equivalence. Approximate expenditure figures remain attributed forecasts. Unknown personal cost does not mean zero.

5. Deployment, reproduction and delivery

Public demo: https://stimmfolgen-demo.stimmfolgen-hosted.workers.dev/ . Phones and computers need no model download, account or payment. A Cloudflare Worker serves the app and calls Public AI's exact swiss-ai/apertus-v1.5-8b route. Provider identity is reported, not weight-attested. The website is globally hosted and is NOT Swiss-sovereign. The owner's Mac is private and never exposed to visitors.

No payment method, paid plan or top-up was added. Existing starter credit supports a finite demo: 200 shared reservations, 80/day, six per five minutes per network, one active generation/audit pair. Durable Object aggregate counts survive restarts; no questions or IPs are persisted by the app. Price ceilings are USD .10/M input and .20/M output; bounded prompts and outputs imply approximately USD .70 total across 200 pairs. Other wallet use can exhaust credit sooner; the credential lasts seven days unless renewed. Provider/platform retention is outside the app's control.

Direct 70B calls returned upstream 429s. The available 8B v5 five-question diagnostic produced one supported answer, one voting-advice abstention and three withheld results. Public HTTPS/browser tests showed a real pension answer with citations, an abstention and honest actor-answer withholding. The phone-sized browser at CSS 389x845 had no horizontal overflow; search, track, dated sources, updates, related vote, perspectives and synthetic encrypted note reload/unlock passed. This is agent-operated browser proof, not physical-phone or independent human-user evidence. Eight hosted tests include actual workerd concurrency and quota persistence using a synthetic provider. A reproduced edge-runtime redirect incompatibility was fixed without allowing credential redirects.

Sovereign deployability is demonstrated by the optional native local route: make local-download then make local-run. Tested on M3 Pro 18 GiB, fresh pinned runtime/source and existing pinned MLX weights, it produced real inference through loopback-only model/app services. A new multi-gigabyte weight download and other hardware were not tested. Root make run still launches non-root read-only Docker reviewed publication with offline evidence replay; it does not run Mac GPU inference inside Linux Docker. Build/setup need internet. Judge instructions and full evidence: docs/JUDGE_RUN.md; docs/evidence/hosted-local-2026-10-15/REVIEW.md.

The new 115-second live walkthrough uses actual public-app captures and local synthetic narration; it is edited footage, not a continuous recording. Public watch page: https://stimmfolgen-demo.stimmfolgen-hosted.workers.dev/demo . The earlier tracker video remains historical. No main merge, organiser submission or terms acceptance is claimed. Third-party source rights remain distinct from Apache-2.0 code and CC-BY-4.0 original report contributions. The eight full-page review images still need redistribution-basis clearance before the organiser warranty; THIRD_PARTY_NOTICES.md also records PyMuPDF AGPL obligations. Original PDF/model weights are not committed.

Next work: second reviewed case, unseen import, language parity, independent semantic/security review and comprehension study. Codex assisted development, tests and documentation; Apertus performed policy inference. References and exact source/model hashes are in the linked evidence manifests and ADRs 0002-0006.